

Lassonde School of Engineering

EECS 2031

Database and BASH

****Bonus Assignment1****

Lab Objectives

To get familiar with cross-referencing data in multiple files using bash and experience working with a database.

Problem Statement

Consider a small database in a bookstore that consists of two files, “books” and “courses”

Books is a file contains records, each record is on a separate line. Each record consists of multiple fields separated by “!” (without the quotes).

Book title, ISBN, Author 1, author 2, Number of copies available in the bookstore.

For example:

The C Programming Language!0–13–110370–9!Brian Kernighan!Dennis Ritchie!44

The courses file contains the following fields

course number, course name, ISBN, R/O

where R/O is a field that contains either “R” if the book is required, or “O” if the book is optional.

For example:

EECS2031!The C Programming Language!0–13–110370–9!R

Write a shell script that runs as “bash 2031A_bonus1.sh flags and arguments”

Depending on the arguments, the script does the following:

- -num followed by a course number. The program displays the textbook title, followed by a “:” then the name of the authors, and the number of copies in the library separated by a single space. For the above record, it should be:

The C programming Language: Brian Kernighan Dennis Ritchie 44

- -nobook without any arguments, the program displays the course number of the courses that does not require a required text. Each course number is on a line by itself.
- -toomuch without any arguments displays the courses that requires more than 2 required textbooks.

Deliverables:

1. **Save as “2031A_bonus1.sh” and submit on eclass in week6 assess module by deadline.**
2. **Demonstrate your work to the instructor during OH, or after class on a Monday by deadline.**

Weightage: 2%

Rubrics:

- **0.5% for** being able to access and cross-reference data in multiple files.
- **0.5% for** implementing -num
- **0.5% for** implementing -nobook
- **0.5% for** implementing -toomuch